LaaS Dashboard

Lab as a Service — Unified Multi-Platform Infrastructure Portal

Version 6.2 · March 2026

Wipro Limited · SDX Infrastructure & Data Centre Operations · Confidential

1. Executive Summary

LaaS (Lab as a Service) is a unified, self-service web portal that consolidates the management of **eight enterprise infrastructure platforms** into a single browser-based interface. Built by the Wipro SDX Infrastructure & Data Centre team, it eliminates the need for engineers and managers to access individual platform consoles, enforcing role-based governance while dramatically accelerating day-to-day operations.

8	1,350+	70+	17
PLATFORMS INTEGRATED	VIRTUAL MACHINES	HOSTS & NODES	DASHBOARD PAGES

Platforms Covered

Platform	Integration Method	Key Capabilities
🖥️ VMware vSphere	vCenter REST API	Full VM lifecycle, snapshots, cloning, vMotion, chargeback
🔴 Red Hat OpenShift	Kubernetes / OCP API	Clusters, nodes, pods, operators, routes, storage, RHOAI
🟩 Nutanix AHV	Prism Central v3 API	Multi-site VMs, hosts, storage, subnets, alerts, provisioning
🤖 Ansible AAP	AAP REST API	Job templates, inventories, credentials, launch, live stdout
☁️ Amazon Web Services	AWS SDK (Boto3)	EC2, S3, RDS, VPCs, Cost Explorer, power actions, SSO tokens
🪟 Microsoft Hyper-V	WinRM PowerShell	VM inventory, power actions, checkpoints, CPU/RAM monitoring
📡 IPAM (SolarWinds)	SolarWinds REST API	Subnet grid, IP utilisation tracking, IP search
📁 Active Directory & DNS	LDAP (MS-ADTS / MS-DNSP)	Users, groups, OUs, DNS zones & records — no WinRM

2. New in v6.2 — Insights & Analytics Engine

The flagship addition in v6.2 is the **Insights & Analytics** page, backed by a **PostgreSQL 16** daily snapshot database. An automated collector runs every night at 11:00 PM, capturing metrics from all eight platforms into twelve dedicated tables with a thirty-day rolling retention window.

🩺 Health Scorecard

- Real-time RAG (Red / Amber / Green) status across all eight platforms
- Trend arrows showing improvement or degradation vs previous day
- Per-platform health detail: CPU%, Memory%, storage, alerts
- Single-glance dashboard for NOC and management reviews

🔀 Change Detection

- Automatic day-over-day drift alerts
- Tracks VM count, host count, pod count, IP utilisation shifts
- Highlights newly added or removed resources
- Colour-coded severity: informational, warning, critical

💰 Capacity & Cost

- Storage and compute exhaustion date predictions
- Monthly chargeback figures in INR (₹) and USD ($)
- Per-platform resource consumption breakdown
- Cost trend over the rolling thirty-day window

📋 Executive Summary

- One-page, PDF-ready overview with aggregate KPIs
- Platform health matrix suitable for management sharing
- Key risk areas and capacity warnings highlighted
- Suitable for weekly status reports and steering committee reviews

📈 Historical Trending & Capacity Forecasting

- Interactive trend charts for 7, 14, and 30-day windows covering all collected metrics
- VM count, host count, pod count, and IP utilisation trends visualised over time
- Linear regression capacity forecasting per metric with exhaustion date predictions
- Storage, compute, and network exhaustion early-warning indicators
- Day-over-day change arrows with percentage shift values
- CSV-exportable historical data snapshots for offline analysis

🗄 PostgreSQL Daily Snapshot Infrastructure

- PostgreSQL 16 database with twelve dedicated snapshot tables
- Automated Task Scheduler job runs daily at 11:00 PM
- Captures: VMware, Hyper-V, Nutanix, OpenShift, AWS, IPAM, AD/DNS, Project Utilisation
- Thirty-day rolling retention with automatic purge of older records
- Retry logic with exponential back-off for resilient multi-platform collection
- Three database views and a purge function for data maintenance

3. Platform Features

VMware vSphere / vCenter

Virtual Machine Management

- Live power state, CPU, RAM & disk across all vCenters
- Power On / Off / Restart from the portal
- Clone VMs and vMotion live migration
- Reconfigure CPU, RAM and disk size dynamically
- Full-text search by name, OS, IP, host
- Bulk VM power and clone actions

Capacity & Host Monitoring

- ESXi host CPU and memory utilisation charts
- Datastore usage with colour-coded free space
- Host ranking — identify most overloaded nodes
- Per-vCenter breakdown for multi-site setups
- VMware resource overview gauges with hover tooltips

Project Chargeback

- VM costs aggregated by VMware tag / project
- Monthly chargeback in INR and USD
- Owner and email mapped to each project tag
- Configurable pricing rates per CPU, RAM and disk
- PDF export for billing and planning

Snapshot Management

- View all snapshots across vCenters
- Create and delete snapshots from portal
- Snapshot age and size tracking
- Identify VMs with excessive snapshots

Red Hat OpenShift

Cluster Topology

- Interactive topology diagram per cluster
- Nodes with CPU/RAM bars and pod counts
- Namespace swim-lanes with health status
- Operator health overlay on topology view

Node Management

- Live nodes with role, status, CPU and RAM
- Cordon, Uncordon and Drain nodes
- Full describe: OS, kernel, kubelet version
- Node age and Kubernetes version display

📦 Pod & Namespace Monitoring

- All pods with namespace, node, phase and age
- Log streaming with container selection
- Namespaces with pod/service/deployment counts
- PVC usage and resource quota visibility

🚀 Routes, Operators, Storage & RHOAI

- All Routes with clickable HTTPS/HTTP URLs
- All cluster operators — Available / Degraded
- Storage classes, provisioner and reclaim policy
- Direct link to RHOAI dashboard per cluster

🟩 Nutanix Prism Central / AHV

🏢 Multi-Site DC / DR

- Separate DC and DR Prism Central grouping
- Add multiple Prism Centrals per site
- Per-PC status indicator and test connectivity
- Live overview KPIs per Prism Central

💻 VMs, Hosts & Provisioning

- All AHV VMs with power state, vCPUs, RAM, IPs
- VM power actions: on, off, reset
- On-demand snapshots via Prism Central API
- 3-step VM provisioning wizard with live image dropdown

🌐 Storage, Networks & Alerts

- Storage containers with usage and free space
- All subnets with VLAN ID and IP pool ranges
- Live alerts with severity: Critical / Warning
- Network type: VLAN or overlay

📊 Capacity & Health Indicators

- Storage total/used with percentage utilisation
- Per-PC cluster health RAG indicators
- CPU and memory utilisation percentages
- Error handling with partial data on timeout

Ansible Automation Platform

Templates & Inventories

- Browse all job templates across AAP instances
- View inventories and credentials
- Launch templates with extra-vars via portal
- Job execution with live stdout streaming

Activity Stream & Audit

- Full automation activity stream per instance
- Filter by event type and timestamp
- Track who launched what and when
- Correlates with the portal-wide audit log

Amazon Web Services

AWS Integration

- EC2 instance discovery with start, stop and reboot actions
- S3 buckets, RDS databases, VPCs and subnet IP utilisation
- IAM user and SSO session token support
- AWS Cost Explorer integration for cost tracking
- Live status indicators in the Overview banner
- Per-instance metadata: type, AZ, public/private IPs

Microsoft Hyper-V

Hyper-V Integration

- Standalone Hyper-V hosts via WinRM PowerShell remoting — no agent required
- VM inventory with power state, CPU, memory and disk
- Power actions: start, stop, restart VMs
- Checkpoint create, restore and delete
- Per-host CPU usage percentage and RAM utilisation bars
- Logical processor count and total/used RAM tracking

📁 Active Directory & DNS

🌲 AD Management

- Browse all AD users with OU and account status
- Create users with password policy & group copy
- Enable / Disable / Unlock / Delete accounts
- Reset passwords, manage group membership
- Pure LDAP integration — no WinRM or PowerShell agent

🌐 DNS Zone & Record Management

- Browse AD-integrated DNS zones
- A, CNAME, MX, PTR, SRV, NS, TXT records
- Add and delete DNS records from the portal
- Flush entire DNS server cache
- MS-DNSP protocol via LDAP — no WinRM required

4. Cross-Platform Capabilities

📡 IPAM & Asset Inventory

- Subnet grid with percentage utilisation and IP counts
- IP search within subnets (SolarWinds integration)
- Top utilised subnets on the Overview page
- Physical/virtual asset tracking with inline editing
- CSV export of asset inventory

🔍 Universal Global Search

- Search VMs, hosts, snapshots, networks, datastores
- OpenShift clusters, Nutanix PCs, Ansible jobs
- Support contacts by name, role, and email
- Keyboard navigation: ↑↓ Enter ESC
- Contact click copies email to clipboard

💬 LaaS AI Assistant

- Natural-language questions about infrastructure
- Voice input — click 🎤 and speak your query
- Powered entirely by live portal data — offline-safe
- Contextually aware of all platforms and metrics

📋 VM Request Workflow

- Single "+ Request VM" button with platform picker
- VMware / OpenShift / Nutanix guided wizards
- Admin-approval workflow with status tracking
- Multi-platform reviews page with platform tabs

💳 Chargeback & Pricing

- Per-platform pricing: VMware, OCP, Nutanix
- Monthly cost breakdown in INR (₹) and USD ($)
- Project-level billing via VMware tags
- Configurable pricing rates saved server-side

🎨 Themes, Charts & Audit

- Three themes: Dark, Light, Wipro Wings (purple)
- Multi-platform donut charts with hover tooltips
- VMware capacity gauges with animated fill bars
- Full portal audit log — every action recorded
- Exportable audit data for compliance

5. Access Control & Roles

Role	Description	Key Permissions
Admin	Full control of all platforms and portal settings	All platform tabs · Add/edit/delete infrastructure sources · Approve/decline VM requests · Manage portal users & roles · Configure pricing, thresholds, notifications · View & export audit log
Operator	Day-to-day operations and provisioning approvals	All VMware VM actions · OCP node cordon/drain · Nutanix VM actions & snapshots · Approve/decline VM requests · AD user/group/DNS management · Launch Ansible templates · View IPAM, Assets, Chargeback
Viewer	Read-only visibility across all platforms	All monitoring tabs (read-only) · IPAM, Assets, Chargeback (read-only) · Ansible: view templates/jobs (no launch) · Full audit log access (view only)
Requester	Self-service VM provisioning requests only	Submit VMware / OpenShift / Nutanix VM requests · Track own request status · View personal audit history · No infrastructure monitoring access

6. Architecture & Technology Stack

Component	Technology	Details
Frontend	React 18 + Vite 7	Single-page app, HTTPS on port 443, served via serve
Backend	Python FastAPI	REST API on port 8001, async endpoints, JWT auth
Database	PostgreSQL 16.6	12 snapshot tables, 3 views, daily auto-purge, port 5433
Daily Collector	Python (Task Scheduler)	Runs at 23:00 daily, 8-platform data capture, retry logic
Authentication	JWT Bearer Tokens	Session-based, role-mapped, stored in sessionStorage
AI Assistant	Local NLP (offline)	No external API calls, fully powered by live portal data
Voice Features	Web Speech API	Text-to-speech voiceover, speech-to-text AI input
Hosting	Windows Server	Services: CaaS-Backend, CaaS-Frontend, CaaS-PostgreSQL

7. Version History

Version	Date	Highlights
v6.2	Mar 2026	Insights & Analytics Engine (4 tabs) · PostgreSQL 16 daily snapshots · Historical Trending · Capacity Forecasting · Daily Collector with retry logic · 30-day rolling retention
v6.1	Mar 2026	AWS EC2/S3/RDS/VPC integration · Hyper-V WinRM integration · Power actions & checkpoints · SSO session tokens · CPU/RAM monitoring
v6.0	Mar 2026	Universal Global Search · Multi-platform donut charts · Revamped Support Escalation Matrix
v5.9	Feb 2026	Ansible Automation Platform · Multi-platform VM Request Wizard (VMware/OCP/Nutanix)
v5.8	Jan 2026	IPAM & Chargeback pages · Subnet utilisation · SolarWinds integration
v5.7	Dec 2025	Nutanix AHV full integration · Multi-site DC/DR · Prism Central v3 API
v5.6	Nov 2025	Dark/Light/Wings themes · Animated ticker · Topology diagrams · AI Chatbot

8. Contact Details

For queries, access requests, or feedback regarding the LaaS Dashboard, please reach out to the team below.

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Distribution: This document is intended for the SDX Infrastructure & Data Centre team, project stakeholders, and management. For access provisioning or demo requests, contact any of the team members listed above.
Dashboard URL: https://<server-ip> (HTTPS, port 443)